TIME-IMM-TSF Extension Methods

Brandi Jones

April 2026

1 Introduction

2 Methods

2.1 Study objective and modeling overview

The goal of this extension was to develop a multimodal predictive framework for intensive care unit (ICU) outcomes using irregular clinical time series and free-text notes from MIMIC, while preserving the motivating structure of irregular time-series learning from the original forecasting framework. We evaluated two binary classification tasks: in-hospital mortality and ICU readmission. The central methodological contribution was a *reliability-aware gated multimodal fusion* module, denoted RA-GRAdd, that modifies the original gate by incorporating modality reliability indicators before fusing structured time-series information with note-derived text information.

Rather than replacing the multimodal fusion idea, this extension preserves its core logic: a structured representation from irregular physiological observations and a text representation from clinical notes are first computed separately, then merged through a learnable gate. The difference is that, in the proposed method, the gate is no longer determined solely by feature content. Instead, it is explicitly adjusted by observed indicators of data reliability, including recency, alignment between note and physiologic timelines, note density, and time-series missingness. This allows the model to learn not only *what* each modality says, but also *how much it should be trusted* for a given admission.

2.2 Data representation

Each admission $i \in \{1, \ldots, n\}$ is represented by three components:

$$\mathbf{x}_i^{(ts)} \in R^{p_{ts}}, \qquad \mathbf{x}_i^{(txt)} \in R^{p_{txt}}, \qquad \mathbf{r}_i \in R^4,$$

where $\mathbf{x}_i^{(ts)}$ is the structured time-series feature vector, $\mathbf{x}_i^{(txt)}$ is the text feature vector, and $\mathbf{r}_i$ is the vector of modality reliability indicators.

The binary outcome is

$$y_i \in \{0, 1\},$$

where $y_i = 1$ indicates the target event, either mortality or ICU readmission depending on the task.

2.2.1 Irregular time-series features

The motivating paper emphasizes irregularly sampled ICU data. To preserve that spirit in a stable classification framework, the structured branch was built from irregular time-series summaries rather than from uniformly resampled measurements. Let admission i have observation times

$$t_{i1} < t_{i2} < \cdots < t_{im_i},$$

with multivariate measurements

$$\mathbf{z}_{ij} \in R^q, \qquad j = 1, \ldots, m_i.$$

From these irregular observations, we constructed features that retain timing and missingness information, including:

$$m_i = numberofobservations,$$

$$\Delta_{ij} = t_{i,j+1} - t_{ij}, \qquad j = 1, \ldots, m_i - 1,$$

$$timespan_i = t_{im_i} - t_{i1},$$

$$\overline{\Delta}_i = \frac{1}{m_i - 1} \sum_{j=1}^{m_i - 1} \Delta_{ij},$$

$$s^2_{\Delta,i} = \frac{1}{m_i - 2} \sum_{j=1}^{m_i - 1} (\Delta_{ij} - \overline{\Delta}_i)^2,$$

and variable-specific summaries such as mean, standard deviation, minimum, maximum, last observed value, linear slope, and missingness rate.

For a single physiological variable k, observed over the irregular sequence as $z_{ij}^{(k)}$, the summary statistics are:

$$\mu_i^{(k)} = \frac{1}{n_i^{(k)}} \sum_{j \in \mathcal{O}_i^{(k)}} z_{ij}^{(k)},$$

$$\sigma_i^{2(k)} = \frac{1}{n_i^{(k)} - 1} \sum_{j \in \mathcal{O}_i^{(k)}} \left(z_{ij}^{(k)} - \mu_i^{(k)} \right)^2,$$

$$z_{i,\min}^{(k)} = \min_{j \in \mathcal{O}_i^{(k)}} z_{ij}^{(k)}, \qquad z_{i,\max}^{(k)} = \max_{j \in \mathcal{O}_i^{(k)}} z_{ij}^{(k)},$$

$$z_{i,last}^{(k)} = z_{i,m_i^{(k)}}^{(k)},$$

where $\mathcal{O}_i^{(k)}$ is the set of observed indices for variable k and $n_i^{(k)} = |\mathcal{O}_i^{(k)}|$.

A simple within-admission linear trend for variable k was computed as the least-squares slope

$$\beta_i^{(k)} = \arg\min_{\beta_0,\beta} \sum_{j \in \mathcal{O}_i^{(k)}} \left(z_{ij}^{(k)} - \beta_0 - \beta j \right)^2,$$

with $\beta_i^{(k)}$ retained as a feature.

The structured vector $\mathbf{x}_i^{(ts)}$ is the concatenation of all such irregularity-aware summaries.

2.2.2 Text features

Let the set of clinical notes for admission i be

$$\mathcal{N}_i = \{d_{i1}, d_{i2}, \ldots, d_{in_i^{(txt)}}\}.$$

A lightweight text representation was used to obtain a robust, reproducible admission-level note vector. In addition to note-count statistics such as number of notes, character count, and word count, a hashed text representation was constructed:

$$\mathbf{x}_i^{(emb)} \in R^{p_{emb}},$$

where each token contributes to a deterministic hashed feature index. The final text vector is

$$\mathbf{x}_i^{(txt)} = \left[\mathbf{x}_i^{(emb)}, n_notes_i, note_char_count_i, note_word_count_i, mean_note_len_i, max_note_len_i \right]^\top.$$

2.3 Reliability indicators

The key methodological extension is the reliability vector

$$\mathbf{r}_i = r_{i,rec} r_{i,align} r_{i,dens} r_{i,miss},$$

which quantifies how trustworthy the modalities are for admission i.

2.3.1 Recency

The recency term was designed to capture whether the structured signal spans the admission with sufficient temporal coverage. In the implemented framework, this term was scaled to reward usable temporal coverage:

$$r_{i,rec} \in [0, 1].$$

Operationally, when the time series contained at least two timestamped observations, the admission received a high recency score. More generally, one can interpret this quantity as a monotone function of usable temporal support:

$$r_{i,rec} = f_{rec}(timespan_i, m_i),$$

with f_{rec} increasing as temporal coverage improves.

2.3.2 Alignment

Alignment measures temporal agreement between the note trajectory and the physiologic trajectory. Let

$$\tilde{t}_i^{(ts)} = \frac{t_{i1} + t_{im_i}}{2}$$

denote the midpoint of the structured time span and let

$$\tilde{t}_i^{(txt)} = \frac{s_{i1} + s_{in_i^{(txt)}}}{2}$$

denote the midpoint of the note timeline, where $s_{i1}, \ldots, s_{in_i^{(txt)}}$ are note timestamps. Then temporal misalignment is

$$\delta_i = \left| \tilde{t}_i^{(ts)} - \tilde{t}_i^{(txt)} \right|,$$

measured in hours, and alignment reliability is

$$r_{i,align} = \exp\left(-\frac{\delta_i}{\tau}\right),$$

where $\tau > 0$ is a decay constant. In the implementation, $\tau = 72$ hours. Thus:

$$r_{i,align} \approx 1$$

when note timing and structured timing are closely centered, and it decreases smoothly toward 0 as they diverge.

2.3.3 Density

Note or measurement density reflects how much information is available per unit time. Let

$$density_i = \frac{m_i}{\max(timespan_i, 1)},$$

where $timespan_i$ is measured in hours. This was clipped into a bounded range:

$$r_{i,dens} = \min(density_i, 1).$$

Admissions with dense irregular sampling receive larger reliability values than admissions with sparse observations.

2.3.4 Missingness

Let the structured observation matrix for admission i be $\mathbf{Z}_i \in R^{m_i \times q}$, with missing indicator matrix

$$M_{ij}^{(k)} = \{\, 1\ , if z_{ij}^{(k)} ismissing, 0, otherwise.$$

The average missingness rate is

$$\bar{M}_i = \frac{1}{m_i q} \sum_{j=1}^{m_i} \sum_{k=1}^{q} M_{ij}^{(k)},$$

and the reliability contribution is

$$r_{i,miss} = 1 - \bar{M}_i.$$

Thus, higher completeness yields greater reliability.

2.4 Original gated multimodal fusion

We first define the original gate used as the baseline. Let

$$\mathbf{x}_i^{(ts)} \in R^{p_{ts}}, \qquad \mathbf{x}_i^{(txt)} \in R^{p_{txt}}.$$

The structured and text branches are projected into a common latent fusion space of dimension d:

$$\mathbf{h}_i^{(ts)} = \phi_{ts}\left(\mathbf{x}_i^{(ts)}\right) = \rho\left(\mathbf{W}_{ts}\mathbf{x}_i^{(ts)} + \mathbf{b}_{ts}\right),$$

$$\mathbf{h}_i^{(txt)} = \phi_{txt}\left(\mathbf{x}_i^{(txt)}\right) = \rho\left(\mathbf{W}_{txt}\mathbf{x}_i^{(txt)} + \mathbf{b}_{txt}\right),$$

where $\rho(\cdot)$ denotes a nonlinear activation, implemented as ReLU followed by dropout.

The original gate is computed only from the feature content:

$$\mathbf{g}_i^{(old)} = \sigma\left(\mathbf{W}_{g2}\,\rho\left(\mathbf{W}_{g1}\mathbf{x}_i^{(ts)}\mathbf{x}_i^{(txt)} + \mathbf{b}_{g1}\right) + \mathbf{b}_{g2}\right),$$

where

$$\mathbf{g}_i^{(old)} \in (0,1)^d$$

is an elementwise gate and $\sigma(\cdot)$ denotes the logistic sigmoid function.

The fused representation is then

$$\mathbf{z}_i^{(old)} = \mathbf{g}_i^{(old)} \odot \mathbf{h}_i^{(ts)} + \left(\mathbf{1} - \mathbf{g}_i^{(old)}\right) \odot \mathbf{h}_i^{(txt)},$$

where $\odot$ is the Hadamard product. This means that each latent dimension is a convex combination of structured and text information:

$$z_{i\ell}^{(old)} = g_{i\ell}^{(old)} h_{i\ell}^{(ts)} + \left(1 - g_{i\ell}^{(old)}\right) h_{i\ell}^{(txt)}, \qquad \ell = 1, \ldots, d.$$

This baseline can learn task-specific modality weighting, but it does not distinguish whether one modality is more trustworthy because of temporal sparsity, missingness, or poor note alignment. It treats content alone as sufficient for fusion.

2.5 Proposed reliability-aware gated residual additive fusion (RA-GRAdd)

2.5.1 Motivation

The original gate can assign high weight to a modality even when that modality is weakly supported by the data. For example, a time-series branch may receive strong weight despite substantial missingness, or a text branch may dominate when notes are temporally poorly aligned with the structured observation window. To address this, we add a second gate component driven by modality reliability.

2.5.2 Reliability projection

Let

$$\mathbf{r}_i \in R^4$$

be the reliability vector. This is projected into the same latent fusion dimension:

$$\mathbf{u}_i = \psi(\mathbf{r}_i) = \mathbf{W}_{r2}\, \rho\left(\mathbf{W}_{r1}\mathbf{r}_i + \mathbf{b}_{r1}\right) + \mathbf{b}_{r2},$$

where

$$\mathbf{u}_i \in R^d.$$

2.5.3 Base gate plus reliability adjustment

A content-based pre-sigmoid gate is first formed:

$$\mathbf{a}_i = \mathbf{W}_{g2}\, \rho\left(\mathbf{W}_{g1}\mathbf{x}_i^{(ts)}\mathbf{x}_i^{(txt)} + \mathbf{b}_{g1}\right) + \mathbf{b}_{g2}.$$

The reliability contribution is then added before sigmoid activation:

$$\mathbf{g}_i^{(RA)} = \sigma\left(\mathbf{a}_i + \mathbf{u}_i\right).$$

Equivalently, for each latent dimension ℓ,

$$g_{i\ell}^{(RA)} = \sigma\left(a_{i\ell} + u_{i\ell}\right).$$

This is the core RA-GRAdd modification. The gate is no longer solely a function of modality content. Instead, it is a *residual additive correction* of the content-based gate by a learned reliability term. This yields a principled interpretation:

- $\mathbf{a}_i$ answers: *what does the content suggest?*
- $\mathbf{u}_i$ answers: *how trustworthy are the modalities?*

The final gate combines both.

2.5.4 Reliability-aware fusion

Using the reliability-aware gate, the fused representation is

$$\mathbf{z}_i^{(RA)} = \mathbf{g}_i^{(RA)} \odot \mathbf{h}_i^{(ts)} + \left(\mathbf{1} - \mathbf{g}_i^{(RA)}\right) \odot \mathbf{h}_i^{(txt)}.$$

At the elementwise level,

$$z_{i\ell}^{(RA)} = g_{i\ell}^{(RA)} h_{i\ell}^{(ts)} + \left(1 - g_{i\ell}^{(RA)}\right) h_{i\ell}^{(txt)}, \qquad \ell = 1, \ldots, d.$$

Because $\mathbf{g}_i^{(RA)}$ depends on reliability, the model can learn patterns such as:

- increase structured weighting when the time series is dense and complete,
- downweight structured information when missingness is severe,
- increase cross-modal trust when note and structured timelines align,
- reduce the influence of poorly aligned or sparse note information.

2.5.5 Comparison of old gate and RA-GRAdd

The baseline gate is

$$\mathbf{g}_i^{(old)} = \sigma(\mathbf{a}_i),$$

while the proposed gate is

$$\mathbf{g}_i^{(RA)} = \sigma(\mathbf{a}_i + \mathbf{u}_i).$$

Therefore, the proposed method can be seen as a strict generalization of the original gate. If

$$\mathbf{u}_i = \mathbf{0},$$

then

$$\mathbf{g}_i^{(RA)} = \mathbf{g}_i^{(old)}.$$

Thus, the original gate is nested within the reliability-aware formulation.

2.6 Binary classifier head

After fusion, the admission-level latent vector $\mathbf{z}_i$ is passed through a multilayer perceptron classifier:

$$\mathbf{m}_i = \rho\left(\mathbf{W}_{c1}\mathbf{z}_i + \mathbf{b}_{c1}\right),$$

$$\eta_i = \mathbf{w}_{c2}^\top \mathbf{m}_i + b_{c2},$$

where η_i is the logit. The predicted event probability is

$$\hat{p}_i = P(y_i = 1 \mid \mathbf{x}_i^{(ts)}, \mathbf{x}_i^{(txt)}, \mathbf{r}_i) = \sigma(\eta_i).$$

For mortality,

$$\hat{p}_i^{(mort)} = \sigma(\eta_i^{(mort)}),$$

and for ICU readmission,

$$\hat{p}_i^{(readm)} = \sigma(\eta_i^{(readm)}).$$

These are trained as separate binary tasks using the same fusion framework.

2.7 Loss function and class imbalance adjustment

Because both tasks are binary and class imbalance can be substantial, training was performed with weighted binary cross-entropy on logits. Let η_i denote the model logit and $\hat{p}_i = \sigma(\eta_i)$. For sample i,

$$\mathcal{L}_i = -w_+ y_i \log(\hat{p}_i) - (1 - y_i) \log(1 - \hat{p}_i),$$

where $w_+ > 0$ is the positive class weight. In implementation,

$$w_+ = \frac{n_-}{n_+},$$

with n_+ and n_- denoting the number of positive and negative examples in the training split.

The empirical objective over the training set is

$$\mathcal{L} = \frac{1}{n_{train}} \sum_{i=1}^{n_{train}} \mathcal{L}_i.$$

The model parameters

$$\Theta = \{\mathbf{W}_{ts}, \mathbf{b}_{ts}, \mathbf{W}_{txt}, \mathbf{b}_{txt}, \mathbf{W}_{g1}, \mathbf{b}_{g1}, \mathbf{W}_{g2}, \mathbf{b}_{g2}, \mathbf{W}_{r1}, \mathbf{b}_{r1}, \mathbf{W}_{r2}, \mathbf{b}_{r2}, \mathbf{W}_{c1}, \mathbf{b}_{c1}, \mathbf{w}_{c2}, b_{c2}\}$$

were estimated by minimizing $\mathcal{L}$ with Adam.

2.8 Decision threshold selection

Although the classifier outputs probabilities, a threshold is needed for binary prediction:

$$\hat{y}_i = I(\hat{p}_i \geq \tau),$$

where $\tau \in (0, 1)$ is the classification threshold and $I(\cdot)$ is the indicator function. Rather than fixing $\tau = 0.5$, the threshold was tuned on the validation set by maximizing the validation F1 score:

$$\tau^* = \arg\max_{\tau \in \mathcal{T}} F_1(\tau),$$

where $\mathcal{T}$ was a grid of candidate thresholds. Final test-set predictions used τ^* selected from the validation data only.

2.9 Ablation study

To determine whether the proposed improvement came from the full reliability formulation or from only one component, an ablation study was performed. Let

$$\mathbf{r}_i = r_{i,rec} r_{i,align} r_{i,dens} r_{i,miss}.$$

The full RA-GRAdd model uses all four terms. In the ablations, one component at a time was removed by setting it to zero before the reliability projection. Formally, let

$$\mathbf{m}^{(a)} \in \{0,1\}^4$$

be an ablation mask. The effective reliability input becomes

$$\tilde{\mathbf{r}}_i^{(a)} = \mathbf{m}^{(a)} \odot \mathbf{r}_i.$$

Then the reliability correction is

$$\mathbf{u}_i^{(a)} = \psi\left(\tilde{\mathbf{r}}_i^{(a)}\right),$$

and the ablated gate is

$$\mathbf{g}_i^{(a)} = \sigma\left(\mathbf{a}_i + \mathbf{u}_i^{(a)}\right).$$

The following models were compared:

1. **Old gate baseline:**
$$\mathbf{g}_i^{(old)} = \sigma(\mathbf{a}_i).$$
2. **Full RA-GRAdd:**
$$\mathbf{g}_i^{(RA)} = \sigma(\mathbf{a}_i + \mathbf{u}_i).$$
3. **RA without recency:**
$$\mathbf{m}^{(-rec)} = [0,1,1,1]^\top.$$
4. **RA without alignment:**
$$\mathbf{m}^{(-align)} = [1,0,1,1]^\top.$$
5. **RA without density:**
$$\mathbf{m}^{(-dens)} = [1,1,0,1]^\top.$$
6. **RA without missingness:**
$$\mathbf{m}^{(-miss)} = [1,1,1,0]^\top.$$

This design allows direct assessment of whether each reliability factor contributes unique predictive value and whether the full reliability-aware gate outperforms both the original gate and each partial variant.

2.10 Interpretability analyses

Several post hoc interpretability analyses were performed.

2.10.1 Gate diagnostics

For each admission, the learned gate vector

$$\mathbf{g}_i \in (0,1)^d$$

was extracted. The mean and standard deviation of each gate dimension across the test set were summarized to examine whether fusion systematically favored structured or text information.

2.10.2 Linear surrogate feature importance

A logistic regression surrogate was fit on the engineered feature table:

$$P(y_i = 1 \mid \mathbf{v}_i) = \sigma(\beta^\top \mathbf{v}_i),$$

where

$$\mathbf{v}_i = \mathbf{x}_i^{(ts)} \mathbf{x}_i^{(txt)} \mathbf{r}_i.$$

Absolute coefficient magnitude

$$|\beta_j|$$

was used as a simple global feature importance measure.

2.10.3 Permutation importance

Permutation importance was computed by randomly permuting a feature v_j across test examples and measuring the corresponding drop in predictive performance. If $S(\cdot)$ denotes a scoring metric such as average precision, then for feature j:

$$PI_j = S(\mathbf{V}, \mathbf{y}) - S(\pi_j(\mathbf{V}), \mathbf{y}),$$

where $\pi_j(\mathbf{V})$ denotes the design matrix after permutation of column j. Larger positive values indicate greater importance.

2.10.4 SHAP analysis

When available, SHAP values were computed on the surrogate linear model. For feature j and sample i, the SHAP contribution is denoted

$$\phi_{ij}.$$

Global importance was summarized as mean absolute SHAP value:

$$SHAP_j = \frac{1}{n_{test}} \sum_{i=1}^{n_{test}} |\phi_{ij}|.$$

2.11 Performance evaluation

Model performance was evaluated on held-out test data using:

$$AUC, \qquad AUPRC, \qquad Accuracy, \qquad F_1, \qquad Precision, \qquad Recall.$$

For a confusion matrix with true positives (TP), false positives (FP), true negatives (TN), and false negatives (FN), the principal threshold-based measures are:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN},$$

$$Precision = \frac{TP}{TP + FP},$$

$$Recall = \frac{TP}{TP + FN},$$

$$F_1 = \frac{2(Precision)(Recall)}{Precision + Recall}.$$

AUC quantifies ranking quality across all thresholds, while AUPRC is especially informative in class-imbalanced clinical prediction problems. Confusion matrices, ROC curves, and precision-recall curves were saved for each experiment.

2.12 Summary of methodological contribution

The methodological contribution of this work is not merely replacing a forecasting head with a classifier. Rather, it is a structured extension of gated multimodal fusion for irregular clinical data. The baseline model learns

$$\mathbf{g}_i^{(old)} = \sigma(\mathbf{a}_i),$$

which is purely content-based. The proposed model instead learns

$$\mathbf{g}_i^{(RA)} = \sigma(\mathbf{a}_i + \mathbf{u}_i),$$

where $\mathbf{u}_i$ is a learned function of modality reliability. This yields a general multimodal classifier that preserves the irregular time-series motivation while making fusion more clinically plausible, because it allows the model to adapt to sparse, misaligned, or incomplete patient records in a principled way.